INCH-POUND
MIL-DTL-85760A (OS)
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SUPERSEDING
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1 August 1986

DETAIL SPECIFICATION

IGNITER, MK 304 MOD 0

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the MK 304 MOD 0 Igniter (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-810	Environmental Test Methods
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

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MIL-STD-1916 DoD Preferred Methods For Acceptance Of Product

DOD-STD-2101 Classification of Characteristics

(Unless otherwise indicated, copies of the above specifications and standards are available from: Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL1638AS106 Igniter, Mk 304 Mod 0

Naval Ordnance Station, Indian Head, Maryland (Code Ident 14083)

DL560-174-001 Test Chamber, Mk 304 Mod 0

(Application for copies should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Documentation Branch (Code 8410), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.9 Sampling Procedures and Tables for Inspection by Variables for Percent Nonconforming

(Application for copies should be addressed to the American Society for Quality, PO Box 3005, 600 N. Plankinton Ave., Milwaukee, WI 53203)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Standard Practice for Radiographic Examination (DoD Adopted)

(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. When specified in the contract or order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Conformance to documents. Igniters shall conform to all requirements of this document and DL1638AS106,

3.3 Radiographic examination (M101). All igniters for the first article test shall be radiograhically examined to determine the presence, location, and condition of components (see 4.5.2).

3.4 Circuit resistance (M102). The igniter circuit resistance, as measured from the connector at 70E " 10EF, shall be between 0.45 and 0.75 ohms (see 4.5.3).

3.5 Ballistic (M103). The igniters shall meet the performance requirements of Table I over the temperature range of !40EF to 140EF (see 4.5.4).

TABLE I. Ballistic performance requirements.

Characteristics[1/]	Specification limit	
	Lower (L)	Upper (U)
Ignition delay (ms)[2/]	-	15
Maximum pressure (psig)[2/]	265	470
Rise time to 140 psig (ms)[2/]	-	25
Pressure-time-interval area (psig-ms)	1240	-

[1/] See 6.4 for definitions
[2/] These characteristics are classified as minor

3.6 Workmanship. Igniters and components shall be free from foreign material, dents, scratches, split or cracked components, poor quality welds, chipping or flaking of painted metal surfaces, sharp edges or other defects or discontinuities as specified on the applicable drawings.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Lot acceptance test (LAT) (see 4.4)

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.3.3 or 4.5.

4.3 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2 and 6.4). First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in production; All parts and materials, including packaging and packing shall be the same as used for production and shall be from the intended sources of supply.

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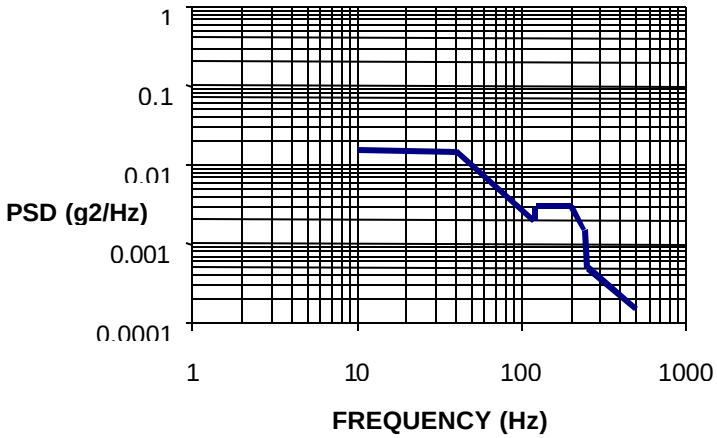
4.3.1 Sample size. Twenty-four igniters shall be submitted for first article inspection.

4.3.2 Inspection routine. The igniters shall undergo the tests specified in Table II in the exact sequence shown.

4.3.3 Test and treatment.

4.3.3.1 Transportation vibration. The igniters shall be subjected to a composite version of MIL-STD-810, Method 514.5, Procedure I, Category 4, Truck Transportation over U.S. Highways, for 3 hours per axis for a total of 9 hours. The composite transportation vibration profile is provided in Figure 1. Each igniter shall be in its shipping configuration for the test. The igniters shall be temperature conditioned to the temperatures specified in Table II for 8 to 9 hours prior to test and maintained at the temperature specified in Table II for the duration of the test.



Breakpoints

Frequency (Hz)	10	40	120	121	200	240	250	500
PSD Value (g2/Hz)	0.01500	0.01500	0.002000	0.00300	0.00300	0.00150	0.00050	0.00015

FIGURE 1. Transportation vibration profile for vertical, longitudinal and transverse axes.

4.3.3.2 Temperature and humidity. Igniters shall be subjected to the temperature and humidity test of MIL-STD-331, Test 105.1, except that the temperature extremes shall be !40°F and +140°F and only one 14-day cycle shall be performed. The igniter shall be tested bare.

4.3.3.3 Drop test. The igniters shall be dropped from 2 feet, measured from the lowest point on the igniter to the impact surface. The impact surface shall be a 2-inch thick steel plate (Brinell hardness of not less than 207) embedded in not less than of 2 feet of reinforced concrete. Two igniters shall be temperature conditioned and maintained at !40EF and two at 140EF. Of these, two (one from each temperature) shall be dropped on the bottom surface of the igniter plug and two (one from each temperature) dropped on the flange of the igniter case.

4.3.4 Failures. Failure of any igniter to pass the first article tests shall be cause for first article rejection.

4.4 Lot acceptance test. Lot acceptance test shall consist of the tests specified in Table III (see 6.2.2).

4.4.1 Inspection lot. An inspection lot shall consist of an igniter lot with no more than one lot of magnesium-fluorocarbon and one lot of initiators.

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TABLE II. First article inspection[1].

Inspection and test	Igniter serial number																								Notes
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Visual	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	see 4.5.1
Circuit resistance	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	see 4.5.3
Radiographic	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	see 4.5.2
Vibration At !40EF													d	d											see 4.3.3.1
At 140EF															d	d									
Temp. and humidity																	d	d	d	d					see 4.3.3.2
2-foot drop At !40EF																					d		d		see 4.3.3.3
At 140EF																						d		d	
Circuit resistance													e	e	e	e	e	e	e	e	e	e	e	e	see 4.5.3
Radiographic													f	f	f	f	f	f	f	f	f	f	f	f	see 4.5.2
Static fire At !40EF	d	d	d	d	d	d							g			g	g	g			g			g	see 4.5.4
At 140EF							d	d	d	d	d	d		g	g				g	g		g	g		

[1] Inspection and tests of each igniter shall be performed in alphabetical order; i.e., the inspection or test opposite the letter “a” shall be performed first, the inspection or test opposite the letter “b” shall be performed second, etc.

TABLE III. Lot acceptance test.

Inspection or test	Req. para.	Method para.
Visual	3.6	4.5.1
Circuit resistance	3.4	4.5.3
Ballistic	3.5	4.5.4

4.4.2 Sampling.

4.4.2.1 For nondestructive tests. Unless otherwise specified in the applicable test method, sampling for and evaluation of nondestructive inspection shall be in accordance with MIL-STD-1916, variable sampling for normal inspection, verification level V, single or double specification limit (as applicable).

4.4.2.2 For destructive tests. The ballistics LAT sample for the igniter lot shall be selected and evaluated in accordance with ANSI/ASQC Z1.9, normal sampling plan, inspection level S4, for each temperature tested. The destructive test shall be evaluated and accepted in accordance with ANSI/ASQC Z1.9, standard deviation method, single or double specification limit (as applicable), variability unknown. In the event of static-firing failure, the sample firings will be terminated for the remaining sample units pending disposition instructions from the procuring activity.

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101 Requirements that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the requirement. Requirements which are not annotated with a classification code are classified minor.

4.5 Tests.

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4.5.1 Visual. The igniter samples shall be visually examined for conformance to DL1638AS106.

4.5.2 Radiographic. The igniters shall be radiographed in accordance with ASTM E1742, radiographic quality 2-1T for conformance to DL1638AS106 (see 3.3).

4.5.3 Circuit resistance. The igniters shall be checked for circuit resistance by applying 0.010 amperes (maximum) to the initiator assembly circuit at 70E " 10EF.

4.5.4 Ballistic test. The igniter samples selected in accordance with 4.4.2.2 shall be static fired (within 10 minutes of removal from conditioning) as specified in 4.4.4.1 through 4.4.4.4.

4.5.4.1 Temperature conditioning.

a. First article inspection. The igniters for first article inspection shall be temperature conditioned to temperatures specified in Table II. The temperature conditioning time shall be 8 to 9 hours prior to test. Igniters not fired within 10 minutes after removal from the conditioning oven shall be reconditioned 2 minutes (over the 10 minutes outside the conditioning chamber). The reconditioning time shall be not more than 8 hours.

b. Lot acceptance test (LAT). Half the igniters for LAT shall be temperature conditioned at !40EF and the other half at 140EF for 8 to 9 hours prior to test. Igniters not fired within 10 minutes after removal from the conditioning oven shall be reconditioned 2 minutes (over the 10 minutes outside the conditioning oven). The reconditioning time shall be not more than 8 hours.

4.5.4.2 Test setup. Igniters shall be fired in a closed chamber conforming to DL560-174-001. Pressure shall be measured through the pressure ports of the closed chamber. The igniters shall be fired by energizing the igniter circuit with 28 volts dc at 7.0 + 0.5, !0.0 amperes.

4.5.4.3 Instrumentation. All measurements shall be made with instruments and systems with a total system accuracy, reproducibility, and frequency response that has been approved by the procuring activity. Calibration records of all test equipment shall be maintained and made available to the procuring activity upon request.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be specified in the contract or purchase order (see 6.2 and 6.5). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Mk 304 Mod 0 igniter is intended for use as the ignition device for the Rocket Motor Mk 117 Mod 0. The igniter is designed for that use only, and thus has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number and date of this specification
b. Whether first article inspection is required and if so, specify the test activity (see 4.3)
c. Production lot size and test activity (see 4.4.1).
d. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M, are applicable
e. Inspection conditions, if other than as specified (see 4.2)
f. Packing and marking requirements, if other than as specified (see 5.1 and 6.5)

6.3 Associated Data Item Descriptions (DIDs). This specification is cited in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), as the source document for the following DIDs. When it is necessary to obtain the data, the applicable DIDs must be listed on the Contract Data Requirements List (DD Form 1423), except where the DoD Federal Acquisition Regulation Supplement exempts the requirement for a DD Form 1423.

DID Number	DID Title
DI-T-4901	First article inspection procedure
DI-T-4902	First article inspection reports
DI-T-4903	Production lot acceptance test (LAT) procedures
DI-T-4904	Production LAT reports
DI-MISC-80043	Ammunition Data Card

The above DIDs were current as of the date of this specification. The current issue of the AMSDL must be researched to ensure that only current and approved DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 24 igniters. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Packaging.

6.5.1 Packaging and packing. The igniter should be packaged and packed in accordance with 49 CFR 171-179 or as specified in the purchase order, requisition, or contract (see 6.2).

6.5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), unit packages and shipping containers should be marked in accordance with MIL-STD-129 and 49 CFR 171-179. All information appearing on the igniter package should be stenciled or otherwise marked on the exterior of the igniter package.

6.5.3 Ammunition data cards. Ammunition data cards, completed in accordance with MIL-STD-1168, should be included with each lot (see 6.3).

6.5.4 Igniter lot numbering system. Lot numbering should be as specified in MIL-STD-1168.

6.6 Definitions. For the purpose of this specification, the following definitions apply.

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6.6.1 Ignition delay. Ignition delay is the time from application of firing current to first indication of pressure on the pressure-time trace, expressed in milliseconds.

6.6.2 Maximum pressure. Maximum pressure is the highest pressure at any point of the pressure-time trace, expressed in psig.

6.6.3 Rise time. Rise time is the time from the application of the firing current to the time pressure reaches 140 psig on the pressure-time trace, expressed in milliseconds.

6.6.4 Pressure-time-interval area. The pressure-time-interval area is the sum of the average pressure multiplied by the time interval for each time increment from 0 to 25 milliseconds on the pressure time trace:

$$pressure\text{ - }time\text{ - }interval\ area = \sum_{0}^{25} (P_{ave} * dt)$$

where: P_{ave} is the average pressure for the given time interval, and dt is the length of the time interval and cannot be greater than 1 millisecond.

6.7 Subject term (key word) listing.

Test chamber, MK 304
Ballistic (M102)
Circuit resistance (M102)

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy – OS
(Project 1340-1032)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-85760A (OS)	2. DOCUMENT DATE (YYMMDD) 030711

3. DOCUMENT TITLE
IGNITER, MK 304 MOD 0

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700	(2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Engineering Documentation Branch (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888	

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